Eiamin Hassan Shanto

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TEACHING INTERESTS

Computer Fundamentals, Structured Programming and Problem Solving, OOPs using Python, Data Structures and Algorithms, Data Communication, Computer Networks, DBMS, Operating System, AI/ML, NLP

EDUCATION

Daffodil International University, Dhaka, Bangladesh 2022 – 2026
Bachelor of Science in Computer Science and Engineering CGPA: 3.91/4.00

Narsingdi Government College, Narsingdi, Bangladesh 2017 – 2019
Higher Secondary Certificate (Science) GPA: 5.00/5.00 (Golden)

Purinda K. M. Sadekur Rahman High School, Narayanganj, Bangladesh 2015 – 2017
Secondary School Certificate (Science) GPA: 5.00/5.00

ACADEMIC EXPERIENCE

Department of Information Technology & Management, Daffodil International University Dhaka, Bangladesh
Graduate Teaching Assistant 2026 – Present

- Assist faculty members in conducting undergraduate theory courses on Engineering Mathematics, Data Communication, Business Mathematics.
- Support course assessment activities including quiz, assignments, and student mentoring.

DIU ACM, Daffodil International University Dhaka, Bangladesh
Mentor Sep 2023 – Dec 2023

- Taught introductory Number Theory and guided problem-solving practice sessions.
- Mentor juniors on basic problem solving classes.
- Conduct viva sessions of Green Division Problem Solvers.

CSE Department, Daffodil International University Dhaka, Bangladesh
Lab Prefect Jul 2023 – Jun 2024

- Assisted students in C programming, debugging, algorithmic thinking, and structured problem-solving during Programming and Problem Solving Lab sessions.
- Guided students in implementing arrays, linked lists, stacks, queues, trees, sorting algorithms, and basic algorithm analysis in DSA lab classes.

DIU Take-Off Programming Contest Dhaka, Bangladesh
Problem Setter 2023

- Set and tested problems for intra-university programming contests.
- Reviewed problem solutions and test cases, and validated programming contest problems through online judge testing.

PROJECTS

Diabetes Prediction Web App: End-to-End Machine Learning Deployment
Project Lead 2026

- Built an end-to-end machine learning web application for diabetes risk prediction using the

Pima Indians Diabetes Dataset.
- Performed exploratory data analysis, including hidden missing-value detection, class distribution analysis, feature correlation study, and age-group risk pattern analysis.
- Trained and compared five classification models: Logistic Regression, Random Forest, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.
- Evaluated model performance using Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, and Cross-Validation.
- Deployed the Streamlit application on Hugging Face Docker Spaces using a custom Dockerfile with containerized dependency management and port configuration.

AI Chatbot
Project Lead 2026
- Built a multi-turn conversational AI chatbot API using FastAPI and Groq LLM with session-based memory and context management.
- Designed RESTful endpoints for chat, history retrieval, and session handling.
- Implemented scalable architecture with Docker and Docker Compose.

RAG Application
Project Lead 2026
- Developing an in-progress Retrieval-Augmented Generation (RAG) application with Python, Streamlit, Google Gemini API, Sentence Transformers, and FAISS for PDF-based question answering.
- Current implementation delivers the core pipeline: PDF ingestion, chunking, embedding-based retrieval, and response generation with configurable environment-driven settings.

Financial Agent
Project Lead 2026
- Developed an AI-powered financial research system integrating Groq LLM, Yahoo Finance, and DuckDuckGo search for stock analysis, market news, and real-time information retrieval.
- Designed a multi-agent workflow to handle stock prices, analyst recommendations, company fundamentals, and finance-related web search queries.
- Built a modular Python-based architecture with API key configuration, testing utilities, and an interactive playground interface running locally for user-friendly financial analysis.

PUBLICATIONS

Conference Publications

- Abdullah Al Noman, **Eiamin Hassan Shanto**, et al. (2025). Advanced Hybrid CNN-ViT Ensemble with Attention and FPN Mechanism for Retinal OCT Disease Classification. [**Published**] International Conference on Intelligent Data Analysis and Applications (IDAA 2025).
- Md. Jobayer Ahmed, Md. Naimul Islam Nuhash, **Eiamin Hassan Shanto**, et al. (2025). Smart Crop Monitoring: Assessing Deep Learning Architectures for Multi-Class Citrus Leaf Disease Recognition. [**Accepted**] International Conference on Intelligent Data Analysis and Applications (IDAA 2025).

AWARDS

4th Place at National Datathon | CUET CSE Fest Chittagong, Bangladesh
Description: Organized by Computer Science & Engineering Department, CUET; Team Name: NeuronX 2025

Champion at Unlock the Algorithm Programming Contest Dhaka, Bangladesh
Description: Organized by Computer Science & Engineering Department, DIU 2023

17th Position at Inter University Programming Contest Dhaka, Bangladesh
Description: Organized by Computer Science & Engineering Department, UAP 2024

Ranked 156th at ICPC Asia Dhaka Regional Preliminary Dhaka, Bangladesh
Description: Team DIU_PigeonHole 2023

EXTRACURRICULAR ACTIVITIES

Participant, AI Innovation Hackathon Dhaka, Bangladesh
Description: Team Leader, AI for Cluster Intelligence Hackathon: Led Cluster-Pulse AI, an ML system that predicts GPU node failures and ranks cost-saving actions. 2026

Session Volunteer, International Conference on Intelligent Data Analysis and Applications (IDAA 2025) Dhaka, Bangladesh
Description: Volunteered at the IDAA 2025 international conference 2025

Technical Volunteer, 2024 ICPC Asia Dhaka Regional Contest Dhaka, Bangladesh
Description: Provided technical support during the regional programming contest 2024

Ranked 2nd in Narsingdi District, Kindergarten Welfare Foundation Scholarship Exam Narsingdi, Bangladesh
Description: District-level scholarship award examination Primary Level

Top 50 Participant at District Science Fair Narsingdi, Bangladesh
Description: Selected among top 50 participants from 150+ schools during my 10th Grade School Level

SKILLS

- **Problem Solving:** Computational Problem Solving, Algorithms, Data Structures, Object-Oriented Programming
- **Programming Languages:** Python, C, C++
- **Database Management:** SQL, MySQL, Database Design, DBMS
- **Research Skills::** Literature Review & Systematic Review, Report Preparation, Experimental Result Analysis
- **Machine Learning & Deep Learning:** Scikit-learn, PyTorch, TensorFlow, NLP, LLM
- **Computer Vision:** OpenCV, CNN, UNet, Vision Transformer
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Version Control:** Git, GitHub
- **Soft Skills:** Academic Writing, Research Communication

REFERENCES

Prof. Dr. Bimal Chandra Das
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Department of Information Technology & Management,
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Dr. S. M. Aminul Haque
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